

DEBRE BERHAN UNIVERSITY
College of Computing
Department of Information System

System analysis and design lecture Note

Chapter1:Introduction to System Development

Data Vs Information

- An information is very important part of any business, which helps the managers to take decisions.
- The data flows throughout the organization and makes it active.
- **The difference between data and information**
 - **Data** are raw(unprocessed) facts about people, objects, events in an organization (Ex. all the sales orders for the year)
 - **Information** is data that have been processed and presented in a form suitable for human interpretation.

Data Vs Information

- An information is very important part of any business, which helps the managers to take decisions.
- The data flows throughout the organization and makes it active.
- **The difference between data and information**
 - **Data** are **raw(unprocessed)** facts about people, objects, events in an organization (Ex. all the sales orders for the year)
 - **Information** is data that have been **processed** and **presented** in a form suitable for human interpretation.

What is a System?

- A collection of **parts** that work together to achieve a common **goal/task/objective**.
- The word system is come from the greek word systema, which means:
 - an organized relationship between among the elements.
 - Those set of elements we call it **subsystems**
 - The irreducible part or an aggregate of parts , also called as a subsystem.
- A collection of **components** that work together to realize some objectives forms a system..

Examples

Information system, Solar system, Digestive systems and transport system etc

What is a System?

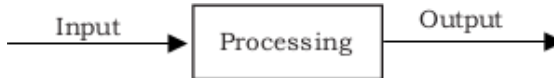
- A collection of **parts** that work together to achieve a common **goal/task/objective**.
- The word system is come from the greek word systema, which means:
 - an organized relationship between among the elements.
 - Those set of elements we call it **subsystems**
 - The irreducible part or an aggregate of parts , also called as a subsystem.
- A collection of **components** that work together to realize some objectives forms a system..

Examples

Information system, Solar system, Digestive systems and transport system etc

cont..

- Basically there are **three major components** in every system, namely input, processing and output.



1

cont..

■ Characteristics of a System

- **Components:** is either an irreducible part or an aggregate of parts , also called as a subsystem
- **Boundary:** A system has boundary, within which all of its components are contained and which establishes the limits of a system, separating it from other systems.
- **Purpose:** All components work together to achieve the overall purpose of the system.
- **Interfaces:** – a points in which which the system meets its environment.
- **Constraints** : are limits to what the system can do (capacity, speed, and capability), ²

cont..

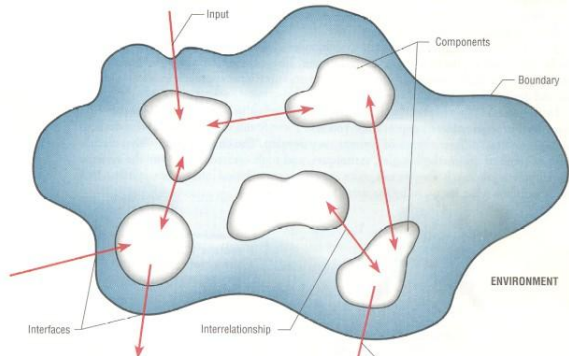
■ Characteristics of a System

- **Components**: is either an irreducible part or an aggregate of parts , also called as a subsystem
- **Boundary**: A system has boundary, within which all of its components are contained and which establishes the limits of a system, separating it from other systems.
- **Purpose**: All components work together to achieve the overall purpose of the system.
- **Interfaces**: – a points in which which the system meets its environment.
- **Constraints** : are limits to what the system can do (capacity, speed, and capability), ²

■ in addition

■ **Environment**

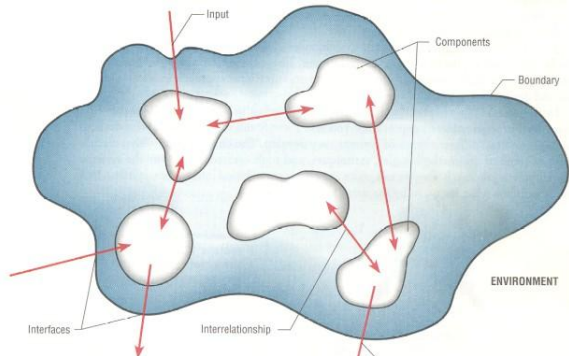
- A system exist within an environment
- everything outside the system's boundary that influences and / or interacts the system.



■ in addition

■ **Environment**

- A system exist within an environment
- everything outside the system's boundary that influences and / or interacts the system.



cont..

What are the basic concepts of a System

■ **Decomposition** :

- is the process of **breaking down** a system into its smaller components.
- Allows the systems analyst to: Break a system into small, manageable subsystems.

■ **Modularity** : is a direct result of decomposition which divides a system into **modules** of a relatively uniform size.

- Modules simplify system design

■ **Coupling** : means that subsystems are **dependent** on each other, messages are passed between subsystems.

- Subsystems that are **dependent** upon each other are **coupled**

■ **Cohesion** : extent to which a subsystem performs single functions.

- modules internal relationship strength(Intra relationship)

cont..

What are the basic concepts of a System

■ **Decomposition** :

- is the process of **breaking down** a system into its smaller components.
- Allows the systems analyst to: Break a system into small, manageable subsystems.

■ **Modularity** : is a direct result of decomposition which divides a system into **modules** of a relatively uniform size.

- Modules simplify system design

■ **Coupling** : means that subsystems are **dependent** on each other, messages are passed between subsystems.

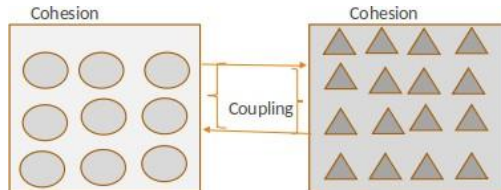
- Subsystems that are **dependent** upon each other are **coupled**

■ **Cohesion** : extent to which a subsystem performs single functions.

- modules internal relationship strength(Intra relationship)

cont..

- Cohesion and Coupling are degrees of **module dependency**.
 - are under **decomposition** concept in **OO software engineering**
 - Cohesion is a measure of the **internal strength** of a module , while
 - coupling is a measure of the extent of information **interchange** between modules .

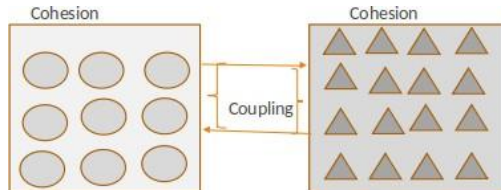


3

³Important System Concepts

cont..

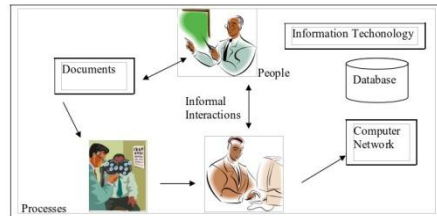
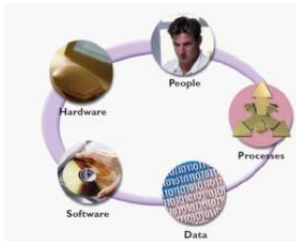
- Cohesion and Coupling are degrees of **module dependency**.
 - are under **decomposition** concept in **OO software engineering**
 - Cohesion is a measure of the **internal strength** of a module , while
 - coupling is a measure of the extent of information **interchange** between modules .



3

What is a Information System?

- **Information System** : is an arrangement of **people, data, processes, interfaces, networks** and technology that interact for the **purpose** of supporting and improving both day-to-day operations in a business.



4

cont..

- Set of **interrelated** components that **collect, process, store** and **distribute data and information** , and provide a feedback mechanism to meet an objective
 - **Hardware** - physical layer of the information system.
 - **Software**- System software's and application software's
 - **Data** - Linked tables work together to supply data.
 - **Processes** - Describe the tasks and business functions that users and managers perform to achieve specific results
 - **People**- Who use the software in order to do their job.
 - Example: **IS specialists, client, clearks**
 - **Controls**- which are the parts of the software written to prevent fraud and theft

cont..

- Set of **interrelated** components that **collect, process, store** and **distribute data and information** , and provide a feedback mechanism to meet an objective
 - **Hardware** - physical layer of the information system.
 - **Software**- System software's and application software's
 - **Data** - Linked tables work together to supply data.
 - **Processes** - Describe the tasks and business functions that users and managers perform to achieve specific results
 - **People**- Who use the software in order to do their job.
 - **Example: IS specialists, client, clerks**
 - **Controls**- which are the parts of the software written to prevent fraud and theft

System development methodologies

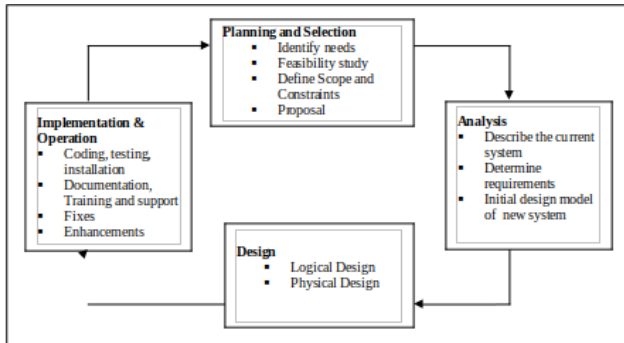
System development life cycle

- series of steps used to mark the phases of development for an information system.
- It is a **common methodology** for systems development.
- **The common four SDLC steps are**
 - 1 Planning and selection
 - 2 Analysis
 - 3 Design and
 - 4 Implementation and operation.
- Some **activities** in one phase in **parallel** with some activities of another phase.
- SDLC life cycle is iterative.
- Each phase has **specific outcomes** and **deliverable** that feed important information to other phase.

cont..

Phases of SDLC

- The SDLC is a structured approach, it uses **data-oriented approach**.



5

cont..

1 systems Planning and Selection : two primary activities

- **Identifying the need for a new or enhanced system**
 - Information needs of the organization are examined and projects to meet these needs are identified from.
 - Requests to deal with problems in current procedures
 - organization may decide whether or not the resources devoted for the project and a careful **feasibility study** is conducted.
- **Investigating the system and determining the proposed system's scope .**
 - Then a specific plan for the proposed project for the team to follow is produced.
 - **Baseline Project Plan** customizes the standardized SDLC.

cont..

1 systems Planning and Selection : two primary activities

- **Identifying the need for a new or enhanced system**
 - Information needs of the organization are **examined** and projects to meet these needs are identified from.
 - Requests to deal with problems in current procedures
 - organization may **decide** whether or not the resources devoted for the project and a careful **feasibility study** is conducted.
- **Investigating the system and determining the proposed system's scope .**
 - Then a specific plan for the proposed project for the team to follow is produced.
 - **Baseline Project Plan** customizes the standardized SDLC.

cont..

2. Systems Analysis

- It has three sub phases:

- First sub phase involves the systems analyst to **determine the requirements of the system**, i.e., what the users want from a proposed system
- Next, the requirements gathered are **structured** (DFD, ERD) according to their interrelationships, eliminating the redundancies
- Third, system analyst has to generate alternative **initial designs** to match the requirements, best **suited design** is selected for the development after the comparison of all alternative designs

cont..

2. Systems Analysis

- It has three sub phases:
 - First sub phase involves the systems analyst to **determine the requirements of the system**, i.e., what the users want from a proposed system
 - Next, the requirements gathered are **structured** (DFD, ERD) according to their interrelationships, eliminating the redundancies
 - Third, system analyst has to generate alternative **initial designs** to match the requirements, best **suited design** is selected for the development after the comparison of all alternative designs

cont..

3. Systems Design

- The system analyst **converts** the description of recommended solution into logical and physical designs
 - Logical design involves in **designing the user interface** , databases and **compute processes**, irrespective of the programming languages (**Algorithms, input and output forms, reports, table normalization**)
 - During the Physical design, the analyst team **decides** the programming language, database systems to be used, hardware platform, operating systems and network environment.
- The final outcome of the design phase is the **physical system specifications**, presented in the form such as a **diagram or written report**

cont..

3. Systems Design

- The system analyst **converts** the description of recommended solution into logical and physical designs
 - **Logical design** involves in **designing the user interface** , databases and **compute processes**, irrespective of the programming languages (**Algorithms, input and output forms, reports, table normalization**)
 - During the Physical design, the analyst team **decides** the programming language, database systems to be used, hardware platform, operating systems and network environment.
- The final outcome of the design phase is the **physical system specifications**, presented in the form such as a **diagram or written report**

cont..

4. Systems Implementation and operation

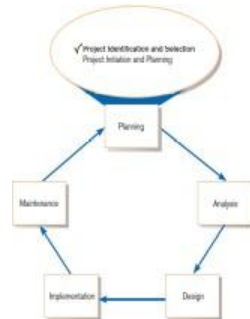
- In this phase the information system is **coded, tested** and **installed** in the organization.
 - Planning for both testing and installation is to be done as early as the project planning and selection phase.
 - This phase also includes the **initial training to the users** and **documentation of the system** documented throughout the life cycle.
- During **operation part**
 - the problems faced by the users should be solved, and changes and enhancements (new versions) is to be made as per the users' desire to reflect changing business conducted.

cont..

Reading assignment

I. System Planning and Selection

- Project Identification and Selection
- Project Initiation and Planning
- Feasibility Study
- Building the Baseline Project Plan
- Reviewing the Baseline Project Plan



cont..

- Deals with the process of **identifying, selecting, initiating, planning** projects and assessing **project feasibility**.

A. Project Identification and Selection

- The first step is to **identify the need** for a system, which can be the result of
 - **Problems** in existing system or process
 - **New feature** required in an existing system
 - **A new idea** for which in Information System is required
 - A requirement to **improve** efficiency in the organization
 - Compulsory standards or **bench marks** by an external organization Ex. Government
 - need to **keep up** with competitors

7

cont..

- Deals with the process of **identifying, selecting, initiating, planning** projects and assessing **project feasibility**.

A. **Project Identification and Selection**

- The first step is to **identify the need** for a system, which can be the result of
 - **Problems** in existing system or process
 - **New feature** required in an existing system
 - **A new idea** for which in Information System is required
 - A requirement to **improve** efficiency in the organization
 - Compulsory standards or **bench marks** by an external organization Ex. Government
 - need to **keep up** with competitors

7

cont..

- During this activity a **senior manager**, a **business group**, an **Information System manager** or a **steering committee** identifies and assess all possible systems development projects.



8

cont..

General Process of Identifying and Selection ISDP

- Process of identifying and selection consists of **three** activities.
 - A. Identifying potential development projects**
 - process may be performed by a key member of **top management**, or a **steering committee** composed of a cross section managers, or User departments, or the development group
 - Projects identified by top management have a **strategic organizational focus**, by the steering committees have a cross functional focus.
 - projects may be **identified** by both top-down and bottom-up initiatives.

cont..

General Process of Identifying and Selection ISDP

- Process of identifying and selection consists of **three** activities.
 - A. Identifying potential development projects**
 - process may be performed by a key member of **top management**, or a **steering committee** composed of a cross section managers, or User departments, or the development group
 - Projects identified by top management have a **strategic organizational focus**, by the steering committees have a cross functional focus.
 - projects may be **identified** by both top-down and bottom-up initiatives.

10

cont..

B. Classifying and ranking IS development projects :done by **top managers, a steering committee**, business units or the IS development group.

The criteria commonly used to evaluate projects

- **Value chain analysis**: Extent to which activities add greatest benefits
- **Strategic alignment**: Extent the projects achieves the long term goals
- **Potential benefits**: Extent to which the project helps to improve profits, Customer service, etc and the duration of the benefits
- **Resource availability**: Amount and type of resources required for the Project
- **Project size / duration**: Number of individuals and duration to complete

cont..

B. Classifying and ranking IS development projects :done by **top managers**, a steering committee, business units or the IS development group.

The criteria commonly used to evaluate projects

- **Value chain analysis**: Extent to which activities add greatest benefits
- **Strategic alignment**: Extent the projects achieves the long term goals
- **Potential benefits**: Extent to which the project helps to improve profits, Customer service, etc and the duration of the benefits
- **Resource availability**: Amount and type of resources required for the Project
- **Project size / duration**: Number of individuals and duration to complete

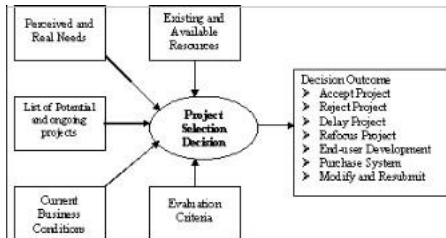
Technical difficulty /

cont..

C. Selecting IS development Project

short and **long** term projects most likely to **achieve** the **business objectives** are considered.

- As business conditions **change over time**, the relative importance of any single project may change.



11

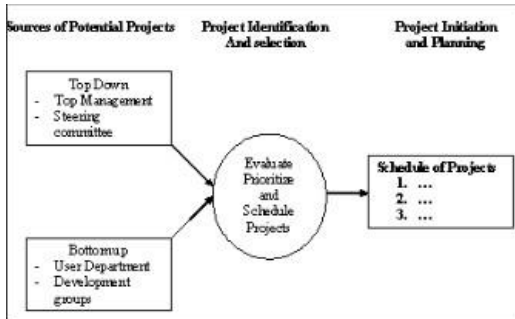
¹¹ Factors to be considered during the project selection

- **factors must be considered** when selecting a project are
 - Perceived needs of the organization
 - Existing systems and ongoing projects
 - Resource availability
 - Evaluation criteria
 - Current business conditions
 - Perspective of the decision makers
- may come from both **top -down** and **bottom-up** sources

- **factors must be considered** when selecting a project are
 - Perceived needs of the organization
 - Existing systems and ongoing projects
 - Resource availability
 - Evaluation criteria
 - Current business conditions
 - Perspective of the decision makers
- may come from both **top -down** and **bottom-up** sources

cont..

- The primary deliverable or end product from the project **identification and selection phase** is a **schedule of specific IS development projects**.



- project move into the second activity called **Project initiation and planning**

cont..

B. Project Initiation and Planning

- objective is to transform a **vague system requirements** into a tangible project description
- Proper project initiation and planning can **reduce** the time consumption of further phases
 - **Project Initiation** : focuses on **activities** that will help to **organize** a team to conduct project planning
 - **Project Planning** : focuses on **defining** clear, discrete tasks and the **work needed** to complete each task
 - The objective of the project planning is to produce two documents : a **Baseline Project Plan (BPP)** and the **Statement of Work (SOW)**.

cont..

B. Project Initiation and Planning

- objective is to transform a **vague system requirements** into a tangible project description
- Proper project initiation and planning can **reduce** the time consumption of further phases
 - **Project Initiation** : focuses on **activities** that will help to **organize** a team to conduct project planning
 - **Project Planning** : focuses on **defining clear, discrete tasks** and the **work needed** to complete each task
 - The objective of the project planning is to produce two documents : **a Baseline Project Plan (BPP)** and the **Statement of Work (SOW)**.

cont..

Baseline project plan

- an internal document used by the **development team** but not shared with customers
- contains all information collected and analyzed during the project initiation and planning activity.
- reflects the best estimate of the **project's scope, benefits, costs, risks** and resource requirements.

statement of the works

- a short document **prepared** for the customers that describes what the project will deliver and outlines all work required to complete the project
- a useful communication tool that **assures** that both system analysts and customers have a common understanding of the project.

■ Example of statement of work

Statement of Work		Prepared : 9/20/2000
Project name:	Customer Tracking Systems	
PVF Project Manager	Jim Woo	
Customer	Marketing	
Project Sponsor	Jackie Hudson	
Project Start / End (Projected)	10/01/00 – 02/01/01	
PVF development staff Estimates (labor months)		
Programmer	2.0	
Jr. Analysts	1.5	
Sr. Analysts	0.3	
Supervisors	0.1	
Consultants	0.0	
Librarian	0.1	
Total	4.0	
Project Description		
Goal		
This project will implement a customer tracking system for the marketing department. The purpose of this system is to automate the ... to save employee time, reduce errors, have more timely information		
Objectives		
- Minimize data entry errors - Provide more timely information ... etc		
Phases of work		
The following tasks and deliverables reflect that current understanding of the project		
In Analysis		
In Design		
In Implementation ...		

cont..

C. Feasibility Study

- Most Information System projects have **budgets** and **deadlines** .
- the **analysis of factors** for feasibility forms the business case (analysis of the assumptions like **resource availability** and potential problems and **system cost and benefits**) that justifies the expenditure of the resources on the project.
- The feasibility factors are in six categories

cont..

■ Economic Feasibility

- Concerned with **assessing** the financial benefits and costs associated with the project.
- To do this, it is necessary to **quantify the monetary** value of the **costs** and **benefits** of the project.
 - a process also called **cost-benefit analysis** .
- Benefits and costs can be tangible or intangible
- **Tangibles** are items which can be quantified in monetary terms and with certainty.
Example . equipment costs, staff/personnel costs, materials costs, conversion costs, training costs.

cont..

■ Economic Feasibility

- Concerned with **assessing** the financial benefits and costs associated with the project.
- To do this, it is necessary to **quantify the monetary** value of the **costs** and **benefits** of the project.
 - a process also called **cost-benefit analysis** .
- Benefits and costs can be tangible or intangible
- **Tangibles** are items which can be quantified in monetary terms and with certainty.
Example . equipment costs, staff/personnel costs, materials costs, conversion costs, training costs.

cont..

- **Intangibles** are items for which a value **cannot** be precisely determined, and where the value may be the result of subjective judgment.

Example . Customer goodwill, employee morale. Operational efficiency

- The sum value of all costs **identified** for the project gives the cost of the system
 - The sum value of all the **benefits** identified for the project gives the benefit of the systems
 - then used to **determine** if the project is economically feasible.
- There are two methods for doing this are **work sheet method** and **present value method** .

cont..

- **Intangibles** are items for which a value **cannot** be precisely determined, and where the value may be the result of subjective judgment.
Example . Customer goodwill, employee morale. Operational efficiency
 - The sum value of all costs **identified** for the project gives the cost of the system
 - The sum value of all the **benefits** identified for the project gives the benefit of the systems
 - then used to **determine** if the project is economically feasible.
- There are two methods for doing this are **work sheet method** and **present value method** .

cont..

■ Operational Feasibility

- a process **examines** whether the new project will attain its desired objectives.
 - to **understand** the degree to which the proposed system will likely **solve** the business problems or the opportunities specified in the Systems requirement documents.

■ Technical Feasibility

- goal of this study is to **understand** the **organization's ability** to construct the proposed system.
- analysis also includes an **assessment of the development group's** understanding of the possible target hardware, software and operating environments as well as the size, complexity and the group's experience with similar systems.

cont..

■ **Schedule Feasibility**

- The process of assessing the degree to which the potential time frame and completion dates for all major activities within a project meet organizational deadlines and constraints for affecting change.

■ **Legal and Contractual Feasibility**

- a process of **assessing** potential legal and contractual ramification due to the construction of a system
- Considerations may include **copyright** or **nondisclosure violation**, **labor laws**, **antitrust legislation**, foreign trade regulations and financial reporting standards as well as current or pending contractual obligations.

- **Political Feasibility** : process of **evaluating** how key stakeholders within the organization view the proposed system.

cont..

D. Building the Baseline Project Plan

- All the information collected during project initiation and planning is collected and organized into a document called the **Baseline Project Plan**.
- Once the BPP is completed, a formal review of the project can be conducted with customers.
 - BPP contains **four major sections**
 - Introduction
 - System Description
 - Feasibility assessment
 - Management issues

cont..

■ Introduction section

- provides a **brief overview** of the entire document and **outline** a recommended course of action for the project
- It provides an **executive summary** that specifies the **project's scope, feasibility**, justification, **resource requirements** and **schedules**.
- Additionally, a **brief statement of the problem**, the environment in which the system is to be implemented and constraints that affect the project are provided.

■ System Description

- provides a **list of alternatives system** configuration.
- It provides a **description** of the selected configuration and a narrative of input information, tasks performed and resultant information.

cont..

■ Feasibility Assessment

- outlines **project costs** and **benefits** and technical difficulties.
- High level **project schedules** are specified using PERT and Gantt charts.
- The greatest amount of project planning effort is typically expended on feasibility assessment activities.

■ Management Issues

- **Team Configuration and Management:** Provides a description of the team member roles and reporting relationships
- **Communication plan:** Provides a description of the **communication procedures** to be followed by management, team members and the customer
- **Project standards and procedures:** Provides a description of how deliverables will be evaluated and accepted by the customer
- **Other project-specific topics:** Provides a description of any **other** Relevant issues related to the project uncovered during planning

cont..

E. Reviewing the Baseline Project Plan

- Before submitting the BPP to some project approval body, it is to be reviewed by the **users, management and development groups**.
- objectives of this review is to **assure** that the proposed system **conforms** to organizational standards and to make sure that all relevant parties understand and agree with the information contained in the BPP.
 - A common method for performing this review is called **a structured walkthrough**, a peer group review of any product created during the systems development process.
 - The walkthrough may have specific agenda that highlights what is to be covered and the expected completion time.
 - In addition to reviewing the BPP, the walkthrough can be used for the following activities
 - System specifications
 - Logical and physical designs
 - Code or program segments
 - Test procedures